

THE ROLE OF INFORMATION SYSTEMS IN PUBLIC REPORTING - A SYSTEMATIC LITERATURE REVIEW

Abstract

Background: *With the advancement of digitalization, information systems are becoming increasingly essential in healthcare, facilitating the collection, analysis, and transformation of data from various sectors to foster availability, accuracy, and timeliness for informed decision-making. Previous studies indicate that public reporting has, overall, contributed to improvements in healthcare quality and facilitated user choice, including enhanced patient satisfaction and clinical outcomes. However, few studies have provided a comprehensive analysis of the effectiveness and limitations of public reporting system.*

Methods: *A literature search was conducted on five electronic databases: PubMed/Medline, PsycInfo, EMBASE, CINAHL, and Web of Science.*

Results: *The review reveals that information systems play a central role in public reporting by supporting data standardization, facilitating cross-system collaboration, and enhancing data usability and accuracy. Patient-centered systems using user-generated data have grown globally since 2015. Key findings highlight the importance of simplified information design and visualization in*

improving user understanding. Furthermore, integrating top-down approach / design with bottom-up clinical engagement—particularly through active participation of healthcare professionals in data collection and indicator development—is critical to strengthening healthcare transparency and quality.

Conclusions: *This systematic literature review examines public reporting from a global perspective, analyzing its evolution, impact, and future potential. By exploring the relationships between public reporting systems, data components, and key stakeholders, this study provides valuable insights into the effectiveness of public reporting in driving healthcare transparency and quality enhancement. The main contributions to this work are as follows: (1) to provide the up-to-date review of public reporting in health care from a global perspective; (2) identifying key trends, challenges, and gaps in digital public reporting systems, summarizing the current state of knowledge.*

Keywords: *Health information system, public reporting, digitalization, health quality, systematic literature review.*

1.Introduction

Public reporting refers to the provision of data on healthcare structures, processes, or outcomes to a broad audience, often free of charge or at symbolic cost, enabling comparisons between different providers or benchmarking against recognized standards or best practices (Totten et al., 2012). One of the earliest known acts of clinical outcomes transparency dates back to 1863, when Florence Nightingale published standardized mortality data from military hospitals, highlighting performance variations across institutions (Dehmer et al., 2014). Modern public reporting systems began in the late 1980s in the United States, when the New York State Department of Health introduced report cards on coronary artery bypass graft (CABG) surgery (Hannan, Kumar, Racz, Siu, & Chassin, 1994). With advancements in technology, information systems have since been adopted in healthcare to support the reporting of health data.

Public reporting within the context of information systems has also garnered increasing attention in research. Previous peer-reviewed studies have explored this topic from the perspectives of individuals, organizations, and public reporting systems. For instance, Saghafian and Hopp (2020) demonstrated that public reporting can enhance patient experiences. Furthermore, studies have

shown that user-friendly presentation formats and accompanying explanations in public reporting improve individuals' understanding and acceptance of quality information (Faber, Bosch, Wollersheim, Leatherman, & Grol, 2009).

From an organizational perspective, public reporting promotes quality management and continuous improvement by enhancing transparency and accountability mechanisms (Cacace, Geraedts, & Berger, 2019). Moreover, incentive measures introduced at the organizational level—such as public reporting and pay-for-performance programs—can motivate healthcare staff and teams to prioritize quality improvement, fostering a culture and environment centered on quality (Anne Frølich, 2007).

In addition, the adoption of digital tools contributes to the reduction of redundant tasks and enhances clinical efficiency. The application of big data and automation technologies in healthcare also demonstrates significant potential in optimizing processes and alleviating human resource constraints (Wang, Kung, & Byrd, 2018).

Notably, some research also examined public reporting in three distinct pathways: the selection pathway, the change pathway and reputation pathway. In the selection pathway, consumers compare public reporting data and choose high-quality providers over low-quality providers (Berwick, James, & Coye, 2003). The selection pathway has been studied in terms of patient choice (Behrendt & Groene, 2016), patient satisfaction (Shintani Smith, Cheng, Kern, Cameron, & Micco, 2021), physician rating website (T. Lagu,

Hannon, Rothberg, & Lindenauer, 2010). In the change pathway, organizations identify underperforming areas, leading to performance improvement. These pathways are interconnected by providers' motivation to maintain or increase market share (Berwick et al., 2003). In the change pathway, studies have examined elements such as transparency mechanisms (Timofeyev, Dremova, & Jakovljevic, 2023) and provider-driven changes, including clinical improvements (Behrendt & Groene, 2016; Berger, Joy, Hutfless, & Bridges, 2013; Campanella et al., 2016) and physician's behavior (Anne Frølich, 2007).

Scholars raised the third pathway, public reporting can negatively affect the public image of a provider or an organization, reputation concerns will therefore motivate providers to protect or improve their public image (Hibbard, Stockard, & Tusler, 2005), thus earn more market share. Although current research accumulate knowledge on public reporting in healthcare, their findings still remain inconsistent, which reflects the complexity of public reporting use in healthcare. Thus, there is a need of systematic literature review to synthesize the understanding of public reporting in information systems in healthcare contexts.

A number of literature reviews have explored public reporting in healthcare, such as investigations into how presentation styles affect consumer decision-making—for example, by simplifying complexity, using non-technical language, or introducing familiar symbols to support information processing (Kurtzman & Greene, 2016). Lagu (2010) conducted an in-depth analysis of 33 physician

rating websites, examining their structure, key features, and the types of information provided, such as physician credentials and patient reviews. In addition, Frølich (2007) proposed an integrated behavioral model that draws on theories from psychology, economics, and organizational behavior to explain how financial and reputational incentives—such as public reporting—can influence clinician behavior individually or in combination. However, they mainly focus on particular type of public reporting, such as public performance reporting or quality reports, or specific user groups, such as clinics, consumers, or governance. There is a lack of extensive understanding of public reporting use among general users in all healthcare contexts. It is particular challenging to understand the roles of information systems in public reporting research.

To fill this research gap, this study aims to conduct a systematic literature review to develop an extensive understanding of the current state of literature on the public reporting in information systems in healthcare contexts and propose potential future research agendas in the field. In order to achieve this research objective, we will focus on address the following research questions:

RQ1 What types of public reporting systems and data components of the public reporting are used and studied?

RQ2 What are the origin countries and key user groups of public reporting systems identified in the reviewed literature?

RQ3 What are the key findings of the reviewed studies on public reporting systems?

Accordingly, this study systematically identifies and reviews the 60 articles on public reporting in information systems in healthcare contexts. The objective of this study is to synthesize existing research on public reporting in the healthcare domain, providing a comprehensive overview and offering recommendations for future research directions for information systems (IS) scholars.

The remainder of this paper is organized as follows: Section 2 outlines the methodology for planning and conducting the literature review. Section 3 presents the findings corresponding to each research question, summarizing the current state of knowledge within an integrated framework reduce the complexity of public reporting by offering structure and clarity. Section 4 outlines potential directions for future research. Section 5 presents the contributions and research implications. Section 6 provides a concluding summary and highlights the limitation.

2. Methods

A research literature review is defined as "a systematic, explicit, and reproducible approach for identifying, evaluating, and synthesizing the existing body of completed and documented work produced by researchers, scholars, and practitioners" (Fink, 2012), when applied rigorously, this method fosters comprehensive synthesis, aiming at providing a clear and reliable foundation for future research and practice. The primary aims of such reviews are to identify emerging research areas, facilitate theory development, address gaps in saturated fields, and categorize studies within specific research streams (Okoli, 2015; Vom Brocke et al., 2015; Watson, 2002). This review follows the guidelines proposed by Okoli (2015).

2.1 Literature search

First, we defined our research purpose is to identify relevant articles on public reporting in information systems in healthcare. Based on the research focus, we developed our search plan and strategy through consulting with the university librarian. We selected five academic databases: PubMed/Medline (National Library of Medicine), PsycInfo (American Psychological Association), EMBASE (Excerpta Medica Database, Elsevier), CINAHL (Cumulative Index to Nursing and Allied Health Literature, EBSCO), and Web of Science. The titles and abstracts of articles were searched using the following terms to identify subject-specific literature

for the review: (“healthcare,” “health care,”) and (“public reporting,”). The search strategy for each database is outlined in Table 1, and the detailed explanation of these search strategies is provided. The search string was designed for broad coverage to avoid missing any relevant research. No limitations were placed on publication date or study design; however, studies published in languages other than English were excluded.

Initially we found 2690 publications in total, of which 746 in Pubmed, 563 in WoS, 411 in CINAHL, 830 in Embase and 140 in PsycInfo. Then we removed 1607 duplicates.

Table 1*Databases and Search Terms*

Database	Search terms
PubMed (746)	("healthcare"[Title/Abstract] OR "health care"[Title/Abstract]) AND "public report*" [Title/Abstract]
PsycInfo (140)	(TI healthcare OR AB healthcare OR TI "health care" OR AB "health care") AND (TI "public report*" OR AB "public report*")
EMBASE (830)	(healthcare OR "health care"):ab,ti AND ("public report*"):ab,ti)
CINAHL (411)	(TI healthcare OR AB healthcare OR TI "health care" OR AB "health care") AND (TI "public report*" OR AB "public report*")
Web of science (563)	(TI=(healthcare OR "health care") OR AB=(healthcare OR "health care")) AND (TI="public report*" OR AB="public report*")

2.2 Article Screening and Selection

In the screening process, first abstracts, keywords, and titles from the five databases of the found articles were screened based on the previously mentioned search terms. Further screening of the 1083 articles was conducted using inclusion and exclusion criteria in a two-step process. In the fourth step, practical screening criteria were applied. The literature review initially screened the titles and abstracts of the 1,083 articles to determine their relevance to public reporting.

After reviewing the titles and abstracts, 993 records were excluded for not matching the inclusion criteria (i.e., studies with the main topic (title/abstract/keywords) containing public report, studies available in English). Our search was confined to English-language publications from 2005 to 2024, as recommended by the Cochrane Rapid Reviews guidance (Garritty et al., 2021). This timeframe was selected based on preliminary searches indicating that the majority of pertinent articles were published during this period.

During the screening process, the titles of the 1,083 retrieved articles were reviewed to determine whether they contained the keyword “public reporting.” If the title is unclear, the abstract was read to assess whether the study focused on public reporting. Articles that merely mentioned public reporting—such as those recommending attention to public reporting or using public reporting data to examine unrelated research—were also excluded.

Citation chaining was conducted to further reinforce the comprehensiveness of the article search. The inclusion criteria and exclusion criteria for each database is summarized in Table 2. By further not fulfilling the inclusion criteria at the analysis level of full text, 8 articles were excluded, leaving 82 studies.

Table 2

Inclusion and Exclusion Criteria

Inclusion criteria (IC)

- IC1: Studies published in English
- IC2: Studies with the main topic (title/abstract) includes public report
- IC3: Publications from 2005 to 2024
- IC4: Full-texts availability of studies
- IC5: Studies published in journals and conferences papers

Exclusion criteria (EC)

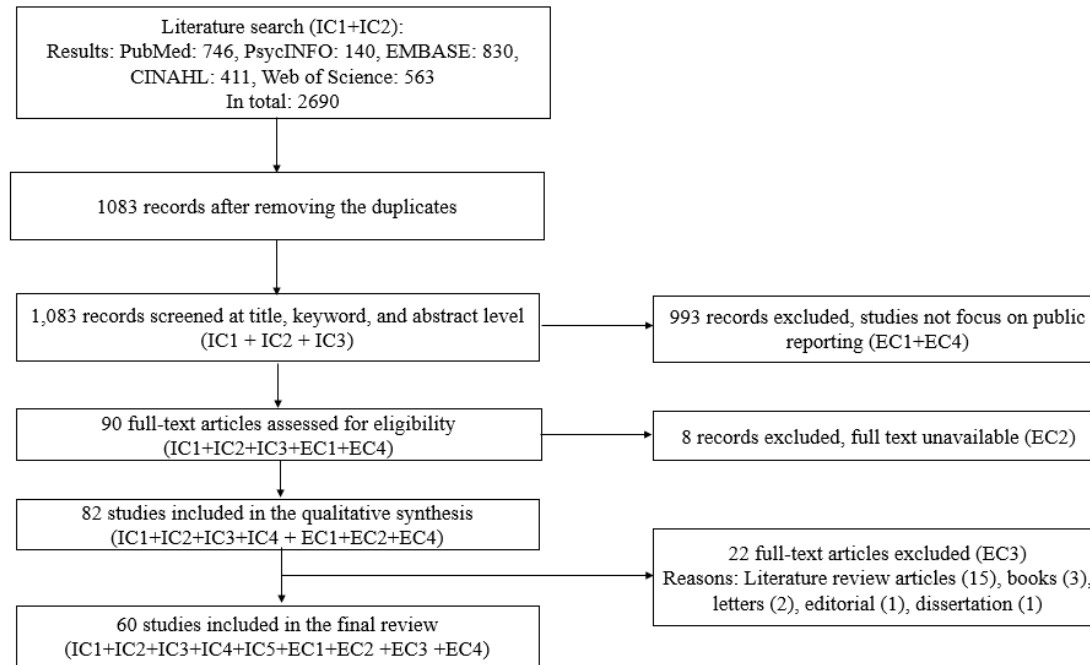
- EC1: Studies whose main topic is not public reporting (i.e., even if public reporting is mentioned in the abstract, the article is not primarily about public reporting)
- EC2: Full text unavailable
- EC3: Studies that are not peer-reviewed articles (books, letters, editorial, dissertations) and literature review articles
- EC4: Publications outside the period of 2005–2024 were excluded

The types of articles were further screened. The remaining 82 articles were reviewed in full to determine whether they met inclusion criterion 5 (i.e., studies published in journals and conference proceedings). Fifteen literature reviews and 8 non-peer-reviewed (e.g., books, editorial, dissertation, letters) were excluded. Conceptual papers published in peer-reviewed journals were included as they contribute valuable theoretical insights. Notably, the excluded articles were organized into a separate subfolder, categorized according to the set inclusion criteria they did not fulfill (Chitu Okoli, 2015).

Three articles were excluded from the analysis of public reporting material because their research focused on general discussions of public reporting without providing any specific data relevant to our analysis. At the end, 60 academic articles were selected as the final sample. Figure 1 illustrates the phases of article retrieval and the number of articles included.

Figure 1

Literature Search Steps



Note. IC: inclusion criterion; EC: exclusion criterion.

2.3 Data Extraction, Coding and Analysis

We aim to provide an overview of research findings on the application of information systems in public reporting within the healthcare domain. Due to the lack of established theoretical frameworks, we adopted an exploratory conventional content analysis

approach (Hsieh & Shannon, 2005). Through the analysis of the included articles, we identified and coded a set of terms or phrases that describe the antecedents and consequences of information systems use in public reporting, based on a thorough reading of the selected studies. We began by documenting the research findings from 60 articles. Based on these findings, we developed preliminary codes. Finally, we conducted a systematic review and synthesis of the codes. The reviewed articles and analysis code are provided in the Appendix.

According to the characteristics of information systems in the context of public reporting, we categorized the codes and established a hierarchical structure consisting of data, information presentation, and policy dimensions to guide the synthesis of the included literature. Specifically, the data and quality metrics primarily focus on improving data accuracy and usability. Regarding information presentation, the format of quality indicator display, the types of indicators valued by patients, and the influence of various demographic factors were all considered. In terms of policy interventions, tailoring strategies to different organizational sizes and contexts facilitates more effective implementation of public reporting. Moreover, combining top-down and bottom-up approaches in policy development was emphasized. Finally, we discussed and confirmed the synthesized findings of this study.

Table 3

Key Themes Coded Based on the Research Questions

Research Questions	Key Themes
(1) What types of public reporting systems and data components of the public reporting are used and studied	• Public reporting systems and data component of the systems based on RQ1
(2) What are the origin countries and key user groups of public reporting systems identified in the reviewed literature	• Key user groups of public reporting systems based on RQ2 • Countries of origins based on RQ2
(3) What are the main findings of the reviewed studies on public reporting systems	• Data and quality metrics based on RQ3 • Presentation and information design based on RQ3 • Policy interventions based on RQ3

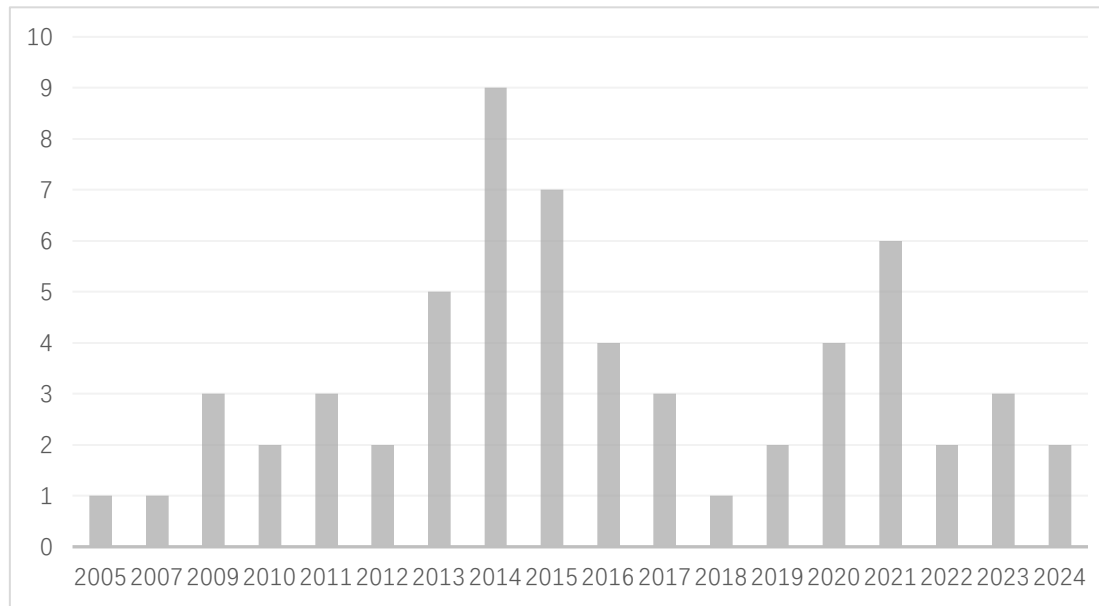
3. Results

3.1 Bibliographical Analysis

In the bibliographical analysis we focused on two key indicators: the country that was object of the reporting, and the number of publications per year worldwide.

Figure 2

Publications per publication year



The data reveal two significant peaks in public reporting research, occurring in 2014 and 2021. The first peak in 2014, with nine published studies, marks a high point in public reporting research. However, from 2015 onward, the number of studies steadily declined. As public reporting transitioned to internet-based platforms, the academic focus on traditional reporting formats diminished, contributing to this decline.

The second peak in 2021 reflects the profound impact of global events, particularly the COVID-19 pandemic, on the academic community. The pandemic renewed interest in healthcare systems and public reporting, leading to a resurgence in research activity.

Between 2023 and 2024, fluctuations in research output may be attributed to shifts in research priorities and resource allocation. Additionally, part of this variation may result from incomplete data, as studies from 2024 may still be in progress or awaiting publication. Overall, this trend highlights the evolution of public reporting research and its dynamic relationship with technological advancements and major global events, offering valuable insights for future studies.

As was shown in Table 4, before year 2015 the number of countries whose public reporting was studied are the United States (n = 25), China (n = 3), Germany (n = 3), Europe (n = 3). After year 2015, the country who studied public reporting becomes to the United States (n = 16), Germany (n = 5), Australia (n = 3), China (n = 1), Canada (n = 1), Belgium (n = 1), Korea (n = 1).

After 2015, countries like Austria, Belgium, and Korea emerged as countries with public reporting studied, indicating that public reporting has been developing globally. In the study by Emmert and Wiener (2017), a comparison was made between the United States and Germany. Although the research was primarily conducted in Germany, leading to the identification of these two countries in this article. Similarly, the study by Du, Zhang, et al. (2015; 2015) evaluated pay-for-performance programs in both the United States and Europe, adding another region to the overall count.

Table 4

Countries Where the Studied Public Reporting Systems were Operating, and Relevant Policy Actions

Country studied	Frequency	Key national policies that might have an effect on publication amounts
United States	41	Affordable Care Act (ACA) (Han & Lee, 2021), pay-for-performance programs (Crews, Chamness, Terry, & Helft, 2016), mandatory vaccination policies (Paris, Arahood, Asche, & Amundson, 2013), mandatory public reporting system for Healthcare-associated infections (HAIs) (Edmond & Bearman, 2007).
Germany	8	Since 2005, German hospitals have been required to regularly publish structured quality reports (Emmert & Wiener, 2017).
Australia	3	The use of pay-for-performance (P4P) (Duncanson et al., 2024).
Europe	3	Healthcare-associated infections (HAIs) (Martin et al., 2013), "patient-centred care" strategies, and pay-for-performance (P4P) approaches (Schlesinger, Grob, & Shaller, 2015).

The bibliographic analysis revealed that the United States (n = 41) and Germany (n = 8) had the highest number of studies. Several articles focused on transparency (in the U.S.) or structured implementation (in Germany), reflecting how policy environments may influence research emphasis. To maintain clarity, Germany is listed separately from studies referencing Europe in general. This categorization follows the source articles and contributes to the consistency of reporting.

3.2 System Types and Their Input and Output

3.2.1 Types of Public Reporting Systems Studied

The term "public reporting" encompasses the dissemination of information and reports regarding the quality of care provided by health and long-term care providers, intended for use by patients and other stakeholders. This information is typically accessible through online platforms, brochures, and other printed materials (Kumpunen, Trigg, & Rodrigues, 2014). Our analysis too indicates that public reporting systems are called with very various names. This also makes their comparison and analysis difficult, as well as finding all relevant articles.

A key distribution goes between digital (computerized) and analog (paper-based) public reporting outputs. By reading the abstracts, introductions, methods and results sections for descriptions, then checking the keywords “internet”, “online”, “web”, “printed”, “paper” and “brochure” in the reviewed article, the majority of public reporting discussed in the reviewed studies are digital (computerized), with the number of 54 (90%). Early studies, particularly those conducted before 2015, focused on analog (paper-based) public reporting, with the number of 3 (5%), which highlights that the Internet has become the predominant medium for studying public reporting. Older studies often centered on analog (paper-based) methods but lacked clarity in their descriptions, whereas newer studies tended to emphasize digital approaches without explicitly detailing their methodologies.

The key names found in our analysis, that we interpret as being at core public reporting systems, even though they might have other functions too, are presented in Table 5.

Table 5

Types of Public Reporting Systems in Reviewed Articles

Type of System	Frequency	Examples of systems	Articles
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Public Reporting Systems	7	• Report cards (e.g., hospitals, nursing homes) • Facebook & Yelp • Ascension Health	(Geraedts, Hermeling, Ortwein, & de Cruppé, 2018), (Emmert & Wiener, 2017), (Emmert, Taheri-Zadeh, Kolb, & Sander, 2017), (Sander, Emmert, Dickel, Meszmer, & Kolb, 2015), (Konetzka, Brauner, Coca Perrailon, & Werner, 2015), (Tara Lagu & Greaves, 2015), (Fakih et al., 2016)
Monitoring and prevention Systems	6	• Healthcare-associated infections (HAIs) • Public Health Automated Case Event Reporting (PACER) • Public health systems	(Martin et al., 2013), (Hebden, 2012), (Mazor, Dodd, & Kunches, 2009), (Edmond & Bearman, 2007), (Mishra, Duke, Lenert, & Karki, 2020), (Chae et al., 2022)
Comparative Reporting Systems	5	• Hospital comparison • Home health comparison • Nursing home comparison • Medical hospital comparison	(Tara Lagu et al., 2019), (Werner, Konetzka, & Polsky, 2016), (Werner et al., 2009), (Emmert, Meier, Pisch, & Sander, 2013), (Emmert & Meier, 2013)
Performance Measurement/Registries	4	• Performance measurement systems • Australia and New Zealand Dialysis and Transplant Registry (ANZDATA) • Canadian Institute for Health Information (CIHI) • National Cardiovascular Data Registry (NCDR)	(Duncanson et al., 2024), (Rambur, Vallett, Cohen, & Tarule, 2013), (Poldrugovac et al., 2022), (Dehmer et al., 2014)

Hospital and Institutional Systems	4	• Hospital Information System (HIS) • National Survey of Substance Abuse Treatment Services (N-SSATS)	(Xin Du, Dan Wang, Xuan Wang, Shiru Yang, & Xinping Zhang, 2015), (Yang, Liu, Wang, Yin, & Zhang, 2014), (Vaughn, Maxey, Keen, Thoele, & Newhouse, 2020), (Saghafian & Hopp, 2020)
Rating Systems	4	• Five-star rating systems • Physician rating websites	(Davila et al., 2024), (Werner et al., 2016), (Emmert et al., 2013), (Emmert & Meier, 2013)
Patient-Centered Systems	3	• Electronic Health Records (EHR) • Electronic Medical Record (EMR)	(Prang et al., 2021), (Tang, Liu, & Zhang, 2017), (Walters et al., 2014)
Claims-Based Systems	3	• All-Payer Claims Databases (APCD) • Wisconsin Collaborative for Healthcare Quality (WCHQ) and Wisconsin Health Information Organization (WHIO)	(Han & Lee, 2021), (Crews et al., 2016), (Toussaint, Shortell, & Mannon, 2014)

The most frequently studied system is the public reporting system ($n = 7$), which includes hospital report cards, provider report cards, and nursing home report cards. Ranked alongside others in second place are the monitoring and prevention system ($n = 6$). Ranked third is the comparative system, which has gained prominence due to the growth of the internet and the expanded use of user-generated data. Research suggests that physician rating websites offer patients a new way to provide feedback and obtain information about physician performance, and their influence will continue to grow as public awareness grows (Emmert et al., 2013). Among these ratings, female physicians are rated significantly higher than male physicians, and older patients and those with private health insurance generally give higher ratings. Most patient reviews tend to be positive, with about two-thirds rating them “very good” (Emmert & Meier, 2013).

With the launch of Nursing Home Compare public reporting, most publicly reported nursing quality indicators have improved (Werner et al., 2009). Public reporting mechanisms, such as Medicare’s Hospital Compare and Physician Compare websites, are driving greater transparency (Tara Lagu et al., 2019). Overall, star rating systems have significantly affected nursing home market share, with consumers preferring higher star-rated nursing homes (Werner et al., 2016).

This trend includes the application of health-related data in various contexts, reflecting the increasing reliance on digital platforms and data-driven decision-making. The plentitude of system types studied under main title “public reporting system” underlines the need for further conceptualization of public reporting systems, a task we too perform for our part in this article.

3.2.2 Key Data Sources and Output Forms for Public Reporting

Previous studies have identified issues such as reporting bias, data distortion (Martin et al., 2013), and misuse due to inaccurate outcome data (Tara Lagu et al., 2019). As any information system, public reporting systems require input data to generate output. This paper classifies studies focusing on performance and quality indicators as input studies since they are conducted before data are released publicly, while studies that analyze publicly available data are categorized as output.

From an input perspective, patient information ($n = 7$) is the primary source, as was shown in Table 6, followed by quality performance data. Research in this area mainly concentrates on quality and performance indicators. Public reporting tends to enhance healthcare quality by drawing the attention of hospital leadership to the data, which in turn encourages hospitals to address the deficiencies highlighted in these reports (Hafner et al., 2011).

In terms of indicator selection, disclosing structural and process measures has proven more effective in improving hospital quality than reporting infection rates directly (Martin et al., 2013). For instance, public reporting has notably improved process measures for acute myocardial infarction (AMI) care, such as timely reperfusion rates, especially in hospitals with limited resources. However, while these process indicators have seen improvements, outcome indicators like mortality rates have not changed significantly, suggesting that further refinement of quality indicators is necessary (Chae et al., 2022).

Table 6

Data (Input) Components of Public Reporting in the Reviewed Articles

Data/Information	Frequency	Input data
Patient-related Data	7	Electronic Health Records (Smith et al., 2023), nursing home report cards (Konetzka et al., 2015), HRC (Hospital Report Cards) (Werner & Asch, 2005), (Emmert & Wiener, 2017), patient demographics (Tang et al., 2017), infection prevention scores (Duncanson et al., 2024), Patient-Reported information (PRI) (Schlesinger et al., 2015).
Quality & Performance Data	6	Risk-Adjusted Outcome Indicators (Bayindir & Schreyögg, 2023), quality indicators (Rambur et al., 2013), patient quality indicators (Ellis et al., 2020), prescription quality indicators (Xin Du, Dan Wang, et al., 2015), five-star rating data (Werner et al., 2016).
Healthcare Institution Data	1	Data from Centers for Medicare and Medicaid Services (CMS) (Saghafian & Hopp, 2020).

As shown in Table 7, most output data in these studies consist of quality information ($n = 7$) and performance information ($n = 6$). With patient-centered system are frequently cited, public reporting developers need to focus more on user experience, ensuring that reporting formats better accommodate diverse user groups (Shaller, Kanouse, & Schlesinger, 2014). Furthermore, research indicates that while patients generally support the public reporting of healthcare performance data, some experience anxiety—especially when they have no option to change treatment centers. Patients prefer that the data be presented positively and accompanied by clear explanations and educational content, which helps them understand the information and make informed decisions (Duncanson et al., 2024).

Table 7

Information (Output) Components of Public Reporting in the Reviewed Articles

Information	Frequency	Examples of data/information
Quality data	7	Acute myocardial infarction (AMI) (Chae et al., 2022), percutaneous Coronary Intervention (PCI) (Wadhera et al., 2019), kidney disease quality data (Duncanson et al., 2024), publicly available performance data (Geraedts et al., 2018; Toussaint et al., 2014), full-time equivalent staff (Han & Lee, 2021), physician information (Geraedts et al., 2018).
Hospital performance data	6	Antibiotic prescription percentage ranking (Xin Du, Dan Wang, et al., 2015), chronic condition management (e.g., diabetes care) (Shaller et al., 2014; Snowden, Kunerth, Carlson, McRae, & Vetta, 2012; Yonek, Mittler, & Hasnain-Wynia, 2014), patient satisfaction scores (Emmert et al., 2017), mortality rates (Crews et al., 2016).
Healthcare Institutions and Care Quality Data	6	CLCs (Community Living Centers) care quality data (Pimentel et al., 2021), PICU (Pediatric Intensive Care Unit) performance data (Natale et al., 2011), Healthcare worker (HCW) flu vaccination coverage (Johnson & Talbot, 2011), prescription behavior data (Xin Du, Dan Wang, et al., 2015), prescription quality indicator data (Xin Du, Xiping Zhang, et al., 2015), prescriber details (name, work unit, time of prescriptions) (Johnson & Talbot, 2011; Martin et al., 2013).
Healthcare-associated infections (HAI)	4	Healthcare-associated infections (HAIs) data (Edmond & Bearman, 2007; Fakih et al., 2016; Hebden, 2012; Mazor et al., 2009).
Operations and Resource Management Data	3	Elective surgery wait times (Prang et al., 2021), operating expenses (Han & Lee, 2021), 30-day readmission rates (Tang et al., 2017), flu vaccination rates (staff behavior data) (Paris et al., 2013).10/7/2026 12:56:00 PM

Public Data	2	Publicly available claims data (Toussaint et al., 2014), HHC (Home Health Compare) public reporting data (K. Jung, Shea, & Warner, 2010), public database systems (e.g., databases maintained by SAMHSA) (Vaughn et al., 2020).
Website information	2	Medicare Nursing Home Compare (Davila et al., 2024), Care website (accessstocare.va.gov) (Werner et al., 2009).
Patient experience data	2	User-generated data (Emmert et al., 2013), patient experience data (Tara Lagu et al., 2019).

3.3 Stakeholders of Public Reporting Systems Identified in the Reviewed Literature

There is an ongoing debate about the effectiveness of existing Health Technology Assessment (HTA) (Fehr et al., 2017). Beyond methodological concerns, discussions emphasize the inclusion and education of key stakeholders and end-users, the adoption of a systematic approach to prioritizing HTA topics, the integration of real-world data, and the establishment of criteria for disinvestment (International Society for Pharmacoeconomics and Outcomes Research, 2016). A systematic analysis of user roles within public reporting systems can help identify diverse needs and examine how these variations influence system development.

In the context of public reporting, user roles are traditionally categorized into purchasers, providers, and regulators (Dehmer et al., 2014). However, the emergence of new user roles, seldom considered in previous studies, led to an inductive approach in this section. Notably, the reviewed articles often discuss multiple user roles, with some employing mixed-methods approaches, thereby incorporating multiple users in a single study.

To provide a comprehensive overview of user engagement in public reporting, we have compiled statistical data on user roles based on their representation in the reviewed studies. As presented in Table 8, findings indicate that staff (n = 50) are the most frequently studied stakeholder group in public reporting research.

Table 8

Classification of Stakeholders in Reviewed Articles

Category	Frequency	Examples of stakeholders
Staff	50	Hospital Administrators (Bayindir & Schreyögg, 2023; Edmond & Bearman, 2007; Ellis et al., 2020; Han & Lee, 2021; Mazor et al., 2009; Natale et al., 2011; Poldrugovac et al., 2022; Werner et al., 2009), Infection Prevention Experts (Hebden, 2012; Martin et al., 2013), Healthcare IT Specialists (Mishra et al., 2020), Nursing Staff (Konetzka et al., 2015; Rambur et al., 2013), Staff responsible for evaluating the performance of hospitals (Paddock, 2014).
Policymakers	33	Public Health Researchers (Chae et al., 2022; Edmond & Bearman, 2007; Ellis et al., 2020; Han & Lee, 2021; Johnson & Talbot, 2011; Tara Lagu et al., 2019; Mazor et al., 2009; Mishra et al., 2020; Poldrugovac et al., 2022; Saghafian & Hopp, 2020; Smith et al., 2023; Van Wilder et al., 2021; Vaughn et al., 2020; Werner et al., 2009), Health Bureau Administrators (Xin Du, Dan Wang, et al., 2015; Prang et al., 2021).
Patients	17	Hospital Patients (J. K. Jung, Wu, Kim, & Polsky, 2016), Hospital Report Cards (HRC) Users (Emmert & Wiener, 2017; Sander et al., 2015), Family Members (J. B. Christianson, Volmar, Alexander, & Scanlon, 2010; Xin Du, Dan Wang, et al., 2015; Konetzka et al., 2015; Tara Lagu & Greaves, 2015; Lihn et al., 2015; Sander et al., 2015; Shaller et al., 2014), Elderly Patients with Chronic Conditions (J. Christianson et al., 2014), Community Patients (J. K. Jung et al., 2016).
Organizations	11	Health Insurance Companies (Negele, Lauerer, Nagel, & Ulrich, 2023; Toussaint et al., 2014; Yang et al., 2014; Yonek et al., 2014), Professional Organizations for Developing and Evaluating Quality Reports (Dehmer et al., 2014; Rambur et al., 2013; Rhoads, Konety, & Dudley, 2009), Health Department (Yang et al., 2014; Yonek et al., 2014), Infection Control Team (Fakih et al., 2016), Canadian Nursing Center Government Regulatory Agency (Poldrugovac et al., 2022).
Family member	8	Patient and family member (J. B. Christianson et al., 2010; Xin Du, Dan Wang, et al., 2015; Konetzka et al., 2015; Tara Lagu & Greaves, 2015; Lihn et al., 2015; Mazor et al., 2009; Sander et al., 2015; Shaller et al., 2014).
Analysts	4	Data Analysts (Crews et al., 2016; Hebden, 2012; Snowden et al., 2012), Public Reporting Developers (Shaller et al., 2014).

The primary aim of public reporting being to aid users in making healthcare decisions, there is limited research focusing on patients ($n = 17$). It is also noteworthy that, at the individual level, patient family members were included in studies, with findings suggesting that their involvement enhances clinical outcomes. However, system use by this role remains underdeveloped.

In the review articles, staff members play a significant role. The employees in these review articles come from various institutions, including hospital nurses, hospital administrators, data analysts, and data collectors from regulatory agencies. Given these diverse roles and institutional backgrounds, the objectivity and subjectivity of their perspectives require further examination and analysis to understand how their positions influence public reporting and decision-making.

3.4 Thematic Overview of the Literature

3.4.1 Data and Quality Metrics

According to the findings in the literature, the main research on data focuses on improving the accuracy and usability of data (Tara Lagu et al., 2019). In terms of improving data accuracy, as stated on the QualityNet website, ensuring data accuracy is crucial for public reporting (Centers for Medicare & Medicaid Services, 2015). The primary research focuses on the rules, standards, and methods of data design (Edmond & Bearman, 2007).

First, in terms of the rules of data design, it is necessary to define data rules in advance before data collection, such as defining differential indicators jointly with the health system to enable clinical system collaboration (Mishra et al., 2020). During the data entry process, strict auditing and verification are carried out to ensure that the entered data truly enhance the quality of patient care, thereby improving the reliability and effectiveness of the data (Crews et al., 2016). Clinical outcome indicators should be clarified to accurately reflect clinical efficacy (Walters et al., 2014). Secondly, standardized data structures can assist in data statistics (Hebden, 2012; Vaughn et al., 2020), for example, through structured clinical documentation.

At the same time, data can be entered in real time during the patient treatment process, and electronic systems can also save staff time (Walters et al., 2014). Finally, supplementing data helps improve the usability of public reporting (Smith et al., 2023). For example, addressing missing data—such as wait times, intake procedures, and current capacity—was helpful for identifying factors which can inform referrals to behavioral health treatment (Lloyd, Cliff, FitzGerald, & Collie, 2021). These studies all indicate that public reporting requires accurate and actionable data to help improve healthcare quality (Toussaint et al., 2014), further combine quality data to develop quality indicator to improve quality (J. B. Christianson et al., 2010).

In terms of collaboration, interaction among individuals, organizations, collaboratives, and systems helps enhance healthcare transparency (Lihn et al., 2015). At the individual level, some studies propose that informed consent forms should include explanatory content to ensure that patient guardians have been provided with data and sufficient explanations to understand the performance of healthcare providers (Lihn et al., 2015). In the design of clinical information systems, more patient decision-making processes should be incorporated, and the roles in these decision processes should also consider community partners, nurses, and doctors.

Shared decision-making can improve actual healthcare quality (Snowden et al., 2012). For example, establishing a unified quality indicator system across patients, nurses, and doctors can better serve patients, professionals, and society. Through various means, this can help in decision-making and quality improvement in long-term care (Davila et al., 2024; Rambur et al., 2013). At the organizational level, employees from different organizations and levels carry out corresponding reporting, such as aligning with system priorities and building inter-system working groups (Smith et al., 2023). There is also research showing that relevant data are collected every six months to monitor various progress in transparency, and that transparent data sharing between centers is allowed; such collaborative methods enhance data usage efficiency (Lihn et al., 2015). Employees are empowered to actively use quality data and integrate it into their daily work to improve actual healthcare quality (Pimentel et al., 2021).

Additionally, healthcare professionals play a very important role in data collection (Natale et al., 2011), and timely, real-time data collection helps enhance data usability. In the collaborative process between healthcare professionals and patients, professionals need to discuss the meaning of the data in public reports with patients and jointly report on the content, such as by providing additional explanations and background information (Duncanson et al., 2024). While building unified data reporting standards, the participation of patients and their families is crucial (Hebden, 2012); not only families, but also community partnerships help improve public reporting awareness and promote equity in health decision-making (Yonek et al., 2014). Finally, some studies have

indicated that establishing dedicated system departments can help alleviate the workload of healthcare institutions and doctors, reducing the burden on physicians (Wadhera et al., 2019), while also increasing the convenience of data access and enhancing the accuracy and reliability of the data (Yang et al., 2014).

3.4.2 Presentation and Information Design

In the selected articles, the presentation of information is another important area of research. In terms of the presentation of information in public reporting, simplified information, explanation and interpretation of indicators, emotionally colored language and context-relevant content, as well as dashboard visualizations, all help improve the presentation of public reports. Some studies have explored the simplification of information in public reporting, such as the use of “star ratings” (Emmert et al., 2017; Emmert & Wiener, 2017). Summary indicators can effectively purify information and enhance the utility of public reports (Werner et al., 2016).

At the same time, the literature supplements the presentation methods, such as using symbol-labeled bar charts, avoiding tables without graphical representation, clearly indicating whether high or low values represent good performance, avoiding

incomplete data, and ranking hospitals based on performance, which helps improve the clarity and understanding of information for the public (Sander et al., 2015). In addition, some studies expanded the approach by providing concise explanations, clarifying center size, disease complexity, geographic and socio-economic context, and using risk-adjusted data with clear explanations of their meanings (Duncanson et al., 2024).

The design of the system should also consider emotional factors, such as using emotionally expressive language and context-related content, which can help relieve consumer anxiety and further promote the development of public reporting (Shaller et al., 2014). Visualization in public reporting helps better display and present information and improves work efficiency for both patients and healthcare providers (Lloyd et al., 2021). One study also tested in rural hospitals whether multiple cross-sectional and temporal measurement studies could be established, and the results improved the presentation effects of public reporting (Lloyd et al., 2021).

Regarding the content of information presentation, public disclosure of hospital and physician quality indicators has a long history. From the patient's perspective, the use of public reporting websites can help reduce the lack of information about physician quality (Emmert et al., 2013). Patients with serious illnesses can benefit from the transparency of information provided in public

reporting (Emmert & Wiener, 2017). However, the continuous development of meaningful performance measurement indicators to help patients understand and use these indicators for decision-making has long been a goal of ongoing academic research (Dehmer et al., 2014). Physicians and patients value different types of information: physicians focus more on data details, professional standards, and treatment outcome indicators of hospitals (Geraedts et al., 2018), while patients value star ratings and narrative information.

Narrative content can enhance personal engagement (Shaller et al., 2014); narrative information integrated with social media use can encourage patients to provide feedback, which reflects a “patient-centered” culture within hospitals (Tara Lagu & Greaves, 2015). However, relying solely on narrative information may introduce bias in patients’ selection of hospitals and doctors (Ellis et al., 2020), such as when patients prioritize reviews and ignore ratings, leading them to choose providers with lower quality scores (Marcotte, Issaka, & Agrawal, 2021). Overall, narrative information can be a supplement information when there is the lack of publicly available public reporting information (Emmert & Meier, 2013). Through feedback on the treatment process and outcomes provided by patients, physicians can better understand the patients’ actual experiences and thoughts and respond constructively (Schlesinger et al., 2015).

Differences among demographic groups are also worth considering. Studies have shown that age, race, education level, and health status all influence public awareness of public reporting. Those with higher education levels tend to have stronger data comprehension and greater willingness to engage than the general public. In contrast, groups with lower educational attainment have a greater need for simplified public reporting information and clear explanations of indicators (Mazor et al., 2009).

In addition, individual information-seeking ability is also required. Some studies suggest that individuals facing critical events or choices are more willing to accept new information, and developers should consider such transitional events to identify target audiences. If consumers are not in a decision-making period, efforts should focus on cultivating their information-searching ability. Different demographic groups have different needs in terms of information-searching capabilities (Shaller et al., 2014). For example, women are generally more active in using online health information, possibly because they care more about family health and are more willing to seek health-related content. Married individuals or those living with a partner are more likely to search for health information online.

Some research has shown that the use of provider rating websites is more common among groups with high healthcare utilization, women, individuals with higher education, and those enrolled in statutory health insurance systems (Emmert et al., 2013). Regarding

differences in health status, it can be reasonably assumed that people with poorer health conditions might use health information websites more frequently. However, multiple studies indicate this is not significantly evident. In fact, some studies have found that individuals in good health are more likely to search for health information online, and those with higher education are more inclined to use online health services (Lee, Jin, Henning-Smith, Lee, & Lee, 2021).

Other research has indicated the effect of overlapping identity traits (J. B. Christianson et al., 2010), such as individuals with both high education and poor health status being more likely to aware physician quality 90dewsinformationio (J. Christianson et al., 2014).

3.4.3 Policy Interventions

According to these studies, the main focus is on the specific implementation effects of public reporting, policy directions, and internal variations. Although different European countries vary greatly in their approaches to publishing infection data, most experts support the public disclosure of such data. They believe that public reporting can positively promote quality-based competition among hospitals, thereby encouraging improvement (Martin et al., 2013). Some studies have mentioned unintended consequences, such as the possibility that physicians might avoid treating critically ill patients to prevent their scores from being negatively affected

(Werner & Asch, 2005). However, research has not found evidence that nursing homes avoid admitting patients with severe dementia (Konetzka et al., 2015). Meanwhile, public reporting has led to improvements in healthcare quality. For instance, a study developed the Infection Composite Score (ICS), which provides a comprehensive assessment tool for infection-related hospitals and helps identify facilities in need of support and improvement (Fakih et al., 2016). In addition, the transparency mechanism of public reporting can improve the rational prescription behavior of primary care physicians (Xin Du, Dan Wang, et al., 2015; Xin Du, Xinpeng Zhang, et al., 2015).

Policy support is another key consideration. Merely increasing quality transparency is not enough; it must be supported by other policy measures to better guide patient choices in healthcare (Saghafian & Hopp, 2020). For example, some studies suggest that combining public disclosure with pay-for-performance mechanisms can create stronger incentives and result in substantial quality improvements (Werner et al., 2009). Other studies use structured consensus-building methods, such as the Delphi method, to define development directions for quality improvement within healthcare systems. Research points out that combining top-down and bottom-up models—for example, establishing quality measurement and public reporting frameworks through top-level design, while giving local regions and stakeholders more autonomy in selective purchasing and payment decisions—should be supported by legislation, with the implementation of clear timelines and monitoring mechanisms (Negele et al., 2023).

Top-down approaches refer to rule and framework formulation led by government or regulatory authorities and implemented through legal measures. Studies have shown that after public reporting programs were implemented in the United States, significant improvements were observed in quality indicators such as bathing, medication management, and mobility. However, emergency medical utilization did not improve, and hospitalization rates even increased, suggesting that overall improvements in service quality are not balanced (J. K. Jung et al., 2016; K. Jung et al., 2010). Moreover, for broader publicly disclosed quality indicators, such as preventable 30-day readmission rates, no improvement was observed—in some cases, outcomes even worsened (Prang et al., 2021). For example, a study on influenza vaccination rates found that mandatory vaccination policies and public reporting were complementary; mandatory policies significantly improved flu vaccination rates among healthcare workers (Johnson & Talbot, 2011).

Bottom-up approaches refer to initiatives led by frontline healthcare organizations and providers who voluntarily participate in defining or optimizing quality indicators. Medical stakeholders, including physicians and nurses, show significant heterogeneity in their views toward existing quality improvement measures (Van Wilder et al., 2021). Meanwhile, individual's participant in public reporting development is helpful to patient centered quality improvement (Hussey, Luft, & McNamara, 2014). Studies indicate that well-resourced tertiary hospitals generally provide higher treatment quality than average institutions. Public reporting highlights

differences among hospitals and emphasizes the need to examine institutional management characteristics and clinical practice standards (Chae et al., 2022). In this process, physicians should also play an active bottom-up role by continuously developing meaningful indicators, participating in data collection, and improving the effectiveness of publicly available data. This can promote the patient-centered treatment in public reporting while minimizing unintended consequences and ensuring the sustainability and connection between quality improvement and reporting (Rambur et al., 2013). It also provides a foundation for integrating public reporting into professional development (Rhoads et al., 2009). Furthermore, one study utilized Bayesian histograms and proposed that policymakers must select appropriate statistical methods when designing payment incentives and public reporting systems, to avoid incorrectly rewarding or penalizing providers (Paddock, 2014).

When developing public reporting policies, it is important to consider not only bottom-up approaches but also the scale of different healthcare institutions and the actual circumstances of various healthcare professionals. By integrating specific market mechanisms, policymakers can stimulate healthy competition among hospitals (Paris et al., 2013). Organizational and employee-level differences also affect the actual outcomes of public reporting policies. Under the pressure from payers, administrators, and peer institutions, public reporting can become more effective (Han & Park, 2021). For instance, in the case of physicians, reputational pressure generated by reporting may motivate behavior change, particularly among those with previously poor

performance, since they face greater external pressure (Tang et al., 2017). Due to the ceiling effect, the impact of performance data disclosure may be limited for high-performing institutions, especially when improvements in certain indicators had already been underway before disclosure (Poldrugovac et al., 2022). Low-quality hospitals are more likely to engage in benchmarking and improvement, while high-quality hospitals tend to rely on maintaining their reputation and altruistic motivations to sustain quality (Bayindir & Schreyögg, 2023). However, this does not mean public reporting data are useless for high-performing hospitals. Some studies suggest that high-performing hospitals are more likely to treat data challenges as opportunities for learning, whereas low-performing hospitals focus more on the timeliness and accuracy of data methodologies (Hafner et al., 2011). Additionally, the intensity of competition can influence the actual implementation of public reporting. In less competitive regions, public reporting policies may improve hospital efficiency. However, in areas where competition is already intense, the marginal benefit of increased transparency is more limited (Han & Lee, 2021).

In summary, when formulating policy, one must consider the varying scales of medical institutions, their differing levels of healthcare quality and performance, and the intensity of local market competition—all of which affect the practical implementation of public reporting. In settings where medical resources are already extremely limited, public reporting policy should also take into account the actual working conditions of healthcare staff.

4. Finding

This study explores the role of information systems in the context of public reporting within the healthcare sector. Drawing on the three pathways through which public reporting promotes healthcare quality—the selection pathway, the change pathway, and the reputation pathway—this review examines the interaction between individuals and systems. Given that public reporting is often policy-driven and executed at the organizational level, research questions have been raised, taking into account system types, countries of origin, policy environments, and primary user groups. The findings underscore the critical role of information systems in supporting data and quality indicators, improving the presentation of information, and facilitating effective policy implementation within public reporting frameworks.

Traditionally, public reporting in healthcare was analog (paper-based). However, since 2015, there has been a significant shift towards digital (computerized). Globally, patient-centered public reporting systems incorporating user-generated data have become prevalent. A coding-based analysis of various studies revealed that countries implementing extensive public reporting policies, such as the United States, also produce the highest volume of related research. However, inconsistencies in national policies lead to significant variations in research backgrounds across different countries. Health technology assessment incorporates distinct traditions, practices, resources, and infrastructure, contributing to a multifaceted and evolving landscape (Fehr et al., 2017).

Existing research primarily focuses on improving the accuracy and usability of data, emphasizing the importance of interactions among individuals, organizations, and systems in improving healthcare transparency. At the system level, establishing standardized data definitions is critical for enabling interoperability across clinical systems. Such standardization contributes to higher data quality and supports more efficient healthcare cost and quality management.

At the individual level, healthcare professionals play a pivotal role in data collection. Patient-related data are the most commonly collected inputs, while data provided by professionals require greater precision. Timely data entry improves both the reliability and practical utility of public reporting. Involving patients in decision-making processes and promoting real-time communication between patients and professionals further improves public understanding of report content and healthcare quality indicators.

At the organizational level, establishing cross-system working groups is an effective strategy for enhancing collaborative efficiency and data transparency. In terms of information presentation, the use of simplified data structures, clear indicator definitions, emotionally resonant language, context-specific content, and visual dashboard tools significantly improves the clarity and effectiveness of public reports.

From a policy perspective, public reporting initiatives are more effective when complemented by supportive policy measures. Successful implementation depends on integrating both top-down strategies (such as regulatory mandates) and bottom-up engagement (such as clinician-led indicator development and data contribution).

Table 9

Main Results Corresponding to Research Questions

Research Questions	Key Findings
(1) What types of public reporting systems and data components of the public reporting are used and studied	• Three main types of systems that are currently most studied in public reporting: public reporting systems, monitoring and prevention systems, and comparative reporting systems. • Input data are primarily patient-related, while output data focus on quality and performance metrics.
(2) What are the origin countries and key user groups of public reporting systems identified in the reviewed literature	• The United States is the most frequently referenced country in the reviewed literature, followed by Australia. • Regarding user groups, the primary users of public reporting systems are organizational staff and policymakers, with patients ranking third.
(3) What are the main findings of the reviewed studies on public reporting systems	• Data: Accuracy and usability of data are core elements that determine the effectiveness of public reporting. • Information presentation: Simplified structures, clear indicator definitions, emotionally resonant language, contextualized content, and dashboard visualizations enhance users' understanding and utilization of public reports. • Policy: Differences in healthcare institution size, service quality levels, and regional market competition significantly affect the practical implementation of public reporting policies.

5. Discussion

Information published by public reporting systems consists of processed data, the processing directed at increasing its usefulness (Ackoff, 1989; Totten et al., 2012). However, public reporting faces several challenges across data quality, performance measurement, and consumer engagement. From a data quality perspective, issues include the reliability of administrative claims data and time lags in public reporting, which diminish its relevance for real-time decision-making (Lihn et al., 2015). Potential biases may arise from manipulation or selective participation, further affecting data integrity (Emmert & Meier, 2013).

In terms of performance data, challenges include inconsistencies in presentation formats, leading to difficulties in consumer interpretation (Hafner et al., 2011). Many consumers struggle to assess multiple provider performance measures without trusted guidance or simplified presentations (Shaller et al., 2014). Additionally, unintended consequences of public reporting, such as poorly adjusted data influencing provider behavior, may lead to risk-averse practices, such as avoiding high-risk patients to maintain favorable metrics (Rhoads et al., 2009).

Report cards introduce another layer of complexity, as providers may selectively report data or avoid high-risk patients to improve their rankings, resulting in biased samples (Werner & Asch, 2005). Furthermore, discrepancies between reported data and actual quality can influence how report cards are perceived and utilized (Pimentel et al., 2021). Data biases have also been noted, particularly in pain assessments for dementia patients, where ascertainment issues affect reporting accuracy (Konetzka et al., 2015).

Electronic health records, despite their value in public reporting, pose challenges related to harmonization, incomplete data, and inconsistent collection methods (Smith et al., 2023). Public health data, while crucial, faces substantial barriers, including a lack of semantic interoperability that hinders data normalization across sources (Mishra et al., 2020). Variability in reporting methodologies across hospitals and countries further complicates benchmarking efforts (Johnson & Talbot, 2011; Martin et al., 2013).

Patient-reported information presents additional obstacles, such as low patient engagement, fragmented data collection, insufficient coordination, and underrepresentation of diverse patient experiences (Tara Lagu & Greaves, 2015). The integration of data from various sources, such as patient narratives, complaints, and patient-reported outcome measures (PROMs), remains a significant challenge (Schlesinger et al., 2015).

Finally, challenges also arise from human factors, including data entry errors and misinterpretation of definitions, necessitating periodic manual corrections that add complexity to the reporting process (Paris et al., 2013). These findings underscore the multifaceted challenges in public reporting, highlighting the need for improved data standardization, risk adjustment, and enhanced usability to support the effectiveness of public reporting in healthcare.

This study systematically identified and reviewed 60 articles on public reporting in healthcare. Over time, public reporting has garnered increasing interest globally, with the Internet becoming the predominant way of developing public reporting, website information has been studied frequently, meanwhile, monitoring and prevention system and the new role of analysis, public reporting gained broad definition. Based on the findings, we will propose future research agendas and discuss research contributions and limitations in the following.

5.1 Future Agenda

5.1.1 Conceptual Agenda

Future research should prioritize developing a comprehensive and interdisciplinary conceptual understanding of public reporting. With the development of the digitalization, the definition of public reporting has expanded. Research in a single discipline led to

the fragmentation within the findings in the literature, hindering the ability to compare and integrate findings effectively. Therefore, further research is required across multiple disciplines, particularly in fields such as health sciences, information systems, psychology, and computer science, to advance understanding and address the complexities of this topic.

5.1.2 *Thematic Agenda*

While previous research has predominantly concentrated on individual-related aspects and organization-related aspects, limited attention has been given to public reporting -related aspects in the healthcare context. Additionally, most studies have primarily examined the motivations behind specific individual side and organization side, with fewer studies addressing the barriers to effective public reporting practices. Moreover, while prior research has explored the outcomes of data usage and standard framework for providing an effective approach to measuring and monitoring health disparities, there remains a notable gap in understanding how data processes function across diverse healthcare contexts to achieve these outcomes. Based on these findings, we propose the following thematic agenda.

First, data serves as a powerful driver of economic growth worldwide (Statista, 2022), among public reporting data, performance data and health records data rank as the most frequently reported categories. However, performance data often suffer from complexity and inconsistent presentation formats, making it difficult for consumers to interpret multiple provider performance metrics—especially in the absence of trusted guidance or simplified visualizations (Shaller et al., 2014). Additionally, publicly reported data, such as infection rates, may be misinterpreted or induce unnecessary patient anxiety, further complicating the effective use of public reporting for informed decision-making (Martin et al., 2013).

Second, there is a need to explore information presentation in the public reporting. Research shows that information displays significantly impact user understanding, for example, bar graphs with symbols, clear markers of high and low values, and hospital rankings based on performance have been shown to enhance patient comprehension of public reporting data (Sander et al., 2015). In advancing personal systems, reports should also place greater emphasis on factors that patients consider significant (Ellis et al., 2020).

Third, there is a need to explore the stakeholder's attitude and actual behavior of public reporting. As digital media capabilities continue to advance, healthcare professionals increasingly require digital health literacy skills to integrate new information into their practices (Slevin et al., 2020). However, many physicians or experts maintain a skeptical or reserved stance on public reporting and use it less frequently (Van Wilder et al., 2021). Given that organizations show improvement following the release of public reporting policy, future research should examine this from an organizational perspective, as organizations function as intermediaries between regulators and individuals. Therefore, future research can focus on the stakeholder's attitude and actual behaviors.

Fourth, there is a need to explore the analysts as a user group. with RQ 1 come up with the monitoring system, analysts had shown up as a new user group in public reporting. Information accessibility and perceived value significantly influence healthcare professionals' behaviors (Xin Du, Xinping Zhang, et al., 2015). Though European physicians primarily fulfill their information needs through online resources (Demergazzi, Pastore, Bassani, Arosio, & Lonati, 2020), they may still rely on other sources for specialized or nuanced information. Monitoring system as new information resources need more study in the future research area.

5.1.3 Technological Agenda

Enhancing system interoperability is another critical focus for technological advancements in public reporting. Improved interoperability can increase data usability, boost operational efficiency, and enable rapid data sharing during infectious disease outbreaks. This helps address issues of data inconsistency across different platforms.

Data standardization and automation have been widely recognized in research as effective ways to improve data accuracy and consistency. Numerous studies emphasize the importance of using standardized definitions and automated systems to ensure reliable public reporting (J. B. Christianson et al., 2010; Crews et al., 2016; Hebden, 2012; Johnson & Talbot, 2011; Mishra et al., 2020; Smith et al., 2023; Vaughn et al., 2020; Walters et al., 2014).

Disparities in data storage systems also present significant challenges. Institutions use different structures—such as relational databases, NoSQL databases, or file systems—which creates fragmentation. Establishing a unified data standard across systems is essential, especially given variations in system updates, as well as differences between organizational and individual data storage practices.

Some studies also highlight significant variation in internal validation methods across healthcare institutions. These differences further underscore the need for consistent standards and validation protocols (Crews et al., 2016).

Finally, the implementation of public reporting systems varies considerably across European countries (Duncanson et al., 2024). This suggests that future research should focus on developing international strategies for data standardization and calibration to support cross-country comparability.

6. Contribution

6.1 Theory Implications

Health information systems play a crucial role in enhancing patient care (Walters et al., 2014). This study contributes to the field by systematically synthesizing and analyzing the current state of research on public reporting systems. First, this study explores how data transmission between systems contributes to data discrepancies.

Second, while pathway theorizing has been extensively developed, the reputation pathway—one of its key components—has received limited attention since 2005. This pathway has the potential to enhance an organization's reputation (Hibbard et al., 2005), thereby increasing market share and fostering a mutually beneficial evolution within the healthcare organization.

Public perception of healthcare quality often diverges from actual clinical performance due to selective reporting and misinformation. By refining the mechanisms through which public reporting influences trust and credibility, healthcare institutions can leverage their reputation not only as a competitive advantage but also as a strategic tool for improving care quality and guiding patient decision-making.

6.2 Practical Implications

First, policymakers should consider market mechanisms when implementing public reporting to ensure optimal healthcare outcomes (Han & Lee, 2021), with incentive structures that particularly address the needs of larger hospitals, as different public reporting programs have varying impacts on clinical quality. Policy should integrate with market mechanism to maximize the effectiveness of public reporting systems (Han & Park, 2021). The degree of improvement varies significantly (Hafner et al., 2011) may be due to increased emphasis on data by hospital leadership. Existing studies highlight the need for incentive policies, which

could effectively enhance quality competition in the healthcare market. Policy making should focus on how these incentives can further encourage high-performing hospitals to improve healthcare outcomes. In this way, a patient-centered public reporting system can be established, enabling the timely collection of patient experiences and outcomes, promoting healthcare transparency.

Second, many practitioners lack formal training in medical coding or documentation improvement, and adherence to standards for physician queries and medical record corrections is suboptimal, with numerous institutions failing to follow prescribed timelines and correction procedures (Crews et al., 2016; Tara Lagu et al., 2019; Martin et al., 2013; Rhoads et al., 2009). By leveraging data for quality improvements, engaging healthcare staff in improvement activities, and clarifying roles and goals, these strategies indirectly contribute to achieving public reporting's quality improvement objectives (Davila et al., 2024).

Third, developers should consider the different stakeholders when developing the system, most public reporting system are directed toward organizational use, patients make up the largest group using public reporting systems, family members of patients also participate in these systems, system development should account for the needs of patient families, particularly in data-sharing options.

7. Conclusions and Limitations

In this paper, a systematic review has been conducted of the literature based on the sample of 60 academic articles. The rise of patient-centered public reporting and user-generated data has expanded the scope of public reporting beyond patients, providers, and regulators to include analysts, monitoring systems, and digital health experts. This shift reflects the broader impact of digitalization in healthcare information dissemination and decision-making.

This study provides an up-to-date and comprehensive review of public reporting in healthcare, identifying key trends, challenges, and future research directions. The findings highlight the importance of standardization, usability, and digital transformation in public reporting. Addressing these challenges can enhance data transparency, stakeholder engagement, and healthcare quality outcomes. Future research should explore interdisciplinary approaches, stakeholder behaviors, and technological advancements to further improve public reporting systems in the healthcare sector.

This systematic literature review has certain limitations. First, as the study focuses specifically on public reporting in healthcare, the search strategy only included studies that explicitly mentioned public reporting. Consequently, relevant research exploring

similar concepts without using this exact term may have been overlooked. Additionally, the review only included studies published in English, potentially omitting valuable insights from research in other languages.

This systematic literature review examines public reporting, analyzing its evolution, impact, and future potential. By exploring the relationships between public reporting systems, data components, and key stakeholders, this study provides valuable insights into the effectiveness of public reporting in driving healthcare transparency and quality enhancement.

References

- Ackoff, R. (1989). From Data to Wisdom.
- Anne Frølich, J. A. T., Peter Broadhead, R. Adams Dudley. (2007). A behavioral model of clinician responses to incentives to improve quality. *Health Policy*, 80. doi:10.1016/j.healthpol.2006.03.001180
- Bayindir, E. E., & Schreyögg, J. (2023). Public Reporting Of Hospital Quality Measures Has Not Led To Overall Quality Improvement: Evidence From Germany: Study examines the impact of public reporting on hospital quality measures in Germany. *Health Affairs*, 42(4), 566–574.
- Behrendt, K., & Groene, O. (2016). Mechanisms and effects of public reporting of surgeon outcomes: a systematic review of the literature. *Health Policy*, 120(10), 1151–1161.
- Berger, Z. D., Joy, S. M., Hutfless, S., & Bridges, J. F. (2013). Can public reporting impact patient outcomes and disparities? A systematic review. *Patient Education and Counseling*, 93(3), 480–487.

- Berwick, D. M., James, B., & Coye, M. J. (2003). Connections between quality measurement and improvement. *Medical care*, 41(1), I-30-I-38.
- Cacace, M., Geraedts, M., & Berger, E. (2019). Public reporting as a quality strategy. *Improving healthcare quality in Europe*, 331.
- Campanella, P., Vukovic, V., Parente, P., Sulejmani, A., Ricciardi, W., & Specchia, M. L. (2016). The impact of public reporting on clinical outcomes: a systematic review and meta-analysis. *BMC Health Services Research*, 16, 1-14.
- Centers for Medicare & Medicaid Services. (2015). *2015 National Impact Assessment of the Centers for Medicare & Medicaid Services (CMS) Quality Measures Report*. Retrieved from <https://www.cms.gov/medicare/quality-initiatives-patient-assessment-instruments/qualitymeasures/downloads/2015-national-impact-assessment-report.pdf>
- Chae, K., Kim, M., Kim, B. O., Jung, C. Y., Kang, H.-J., Oh, D.-J., . . . Han, K.-R. (2022). Public Reporting on the Quality of Care in Patients with Acute Myocardial Infarction: The Korean Experience. *International journal of environmental research and public health*, 19(6), 3169.
- Chitu Okoli, K. S. (2015). A Guide to Conducting a Systematic Literature Review of Information Systems Research.
- Christianson, J., Maeng, D., Abraham, J., Scanlon, D. P., Alexander, J., Mittler, J., & Finch, M. (2014). What influences the awareness of physician quality information? Implications for Medicare. *Medicare & medicaid research review*, 4(2), mmrr. 004.002. a002.
- Christianson, J. B., Volmar, K. M., Alexander, J., & Scanlon, D. P. (2010). A report card on provider report cards: current status of the health care transparency movement. *Journal of general internal medicine*, 25, 1235-1241.
- Crews, H. R., Chamness, A. R., Terry, C. L., & Helft, P. R. (2016). The use of claims-based data in inpatient public reporting and pay-for-performance programs: is there opportunity for improvement? *The Journal for Healthcare Quality (JHQ)*, 38(6), 379-395.
- Davila, H., Mills, W. L., Clark, V., Hartmann, C. W., Sullivan, J. L., Mohr, D. C., . . . Pimentel, C. B. (2024). Quality Improvement Efforts in VA Community Living Centers Following Public Reporting of Performance. *Journal of aging & social policy*, 36(1), 118-140.

- Dehmer, G. J., Drozda, J. P., Brindis, R. G., Masoudi, F. A., Rumsfeld, J. S., Slattery, L. E., & Oetgen, W. J. (2014). Public reporting of clinical quality data: an update for cardiovascular specialists. *Journal of the American College of Cardiology*, 63(13), 1239–1245.
- Demergazzi, S., Pastore, L., Bassani, G., Arosio, M., & Lonati, C. (2020). Information needs and information-seeking behavior of Italian neurologists: exploratory mixed methods study. *Journal of medical Internet research*, 22(4), e14979.
- Du, X., Wang, D., Wang, X., Yang, S., & Zhang, X. (2015). Exploring the transparency mechanism and evaluating the effect of public reporting on prescription: a protocol for a cluster randomized controlled trial. *BMC Public Health*, 15, 277. doi:10.1186/s12889-015-1454-6
- Du, X., Wang, D., Wang, X., Yang, S., & Zhang, X. (2015). Exploring the transparency mechanism and evaluating the effect of public reporting on prescription: a protocol for a cluster randomized controlled trial. *BMC Public Health*, 15, 1–9.
- Du, X., Zhang, X., Tang, Y., Zhang, X., Wang, L., & Wang, X. (2015). How public reporting of prescription quality indicators influence prescribing practices? A survey of general practitioners. *Journal of Evaluation in Clinical Practice*, 21(5), 943–951.
- Duncanson, E., Davies, C. E., Muthuramalingam, S., Johns, E., McColm, K., Hempstalk, M., . . . McDonald, S. P. (2024). Patient perspectives of center-specific reporting in kidney failure care: an Australian qualitative study. *Kidney International Reports*, 9(4), 843–852.
- Edmond, M. B., & Bearman, G. M. (2007). Mandatory public reporting in the USA: an example to follow? *Journal of Hospital Infection*, 65, 182–188.
- Ellis, R. J., Yuce, T. K., Hewitt, D. B., Merkow, R. P., Kinnier, C. V., Johnson, J. K., & Bilimoria, K. Y. (2020). National evaluation of patient preferences in selecting hospitals and health care providers. *Medical care*, 58(10), 867–873.
- Emmert, M., & Meier, F. (2013). An analysis of online evaluations on a physician rating website: evidence from a German public reporting instrument. *Journal of medical Internet research*, 15(8), e2655.
- Emmert, M., Meier, F., Pisch, F., & Sander, U. (2013). Physician choice making and characteristics associated with using physician-rating websites: cross-sectional study. *Journal of medical Internet research*, 15(8), e2702.

- Emmert, M., Taheri-Zadeh, F., Kolb, B., & Sander, U. (2017). Public reporting of hospital quality shows inconsistent ranking results. *Health Policy, 121*(1), 17–26.
- Emmert, M., & Wiener, M. (2017). What factors determine the intention to use hospital report cards? The perspectives of users and non-users. *Patient Education and Counseling, 100*(7), 1394–1401.
- Faber, M., Bosch, M., Wollersheim, H., Leatherman, S., & Grol, R. (2009). Public reporting in health care: How do consumers use quality-of-care information?: A systematic review. *Medical care, 47*(1), 1–8.
- Fakih, M. G., Skierczynski, B., Bufalino, A., Groves, C., Roberts, P., Heavens, M., . . . Haydar, Z. (2016). Taking advantage of public reporting: An infection composite score to assist evaluating hospital performance for infection prevention efforts. *American Journal of Infection Control, 44*(12), 1578–1581.
- Fehr, R., Alexanderson, K., Favaretti, C., De Jong, J., La Torre, G., Lim, T.-A., . . . Rosenkötter, N. (2017). Health assessments for health governance—concepts and methodologies. *The European Journal of Public Health, 27*(4), 609–616.
- Fink, A. (2012). *Evidence-based public health practice*: Sage publications.
- Garritty, C., Gartlehner, G., Nussbaumer-Streit, B., King, V. J., Hamel, C., Kamel, C., . . . Stevens, A. (2021). Cochrane Rapid Reviews Methods Group offers evidence-informed guidance to conduct rapid reviews. *Journal of clinical epidemiology, 130*, 13–22.
- Geraedts, M., Hermeling, P., Ortwein, A., & de Cruppé, W. (2018). Public reporting of hospital quality data: What do referring physicians want to know? *Health Policy, 122*(11), 1177–1182.
- Hafner, J. M., Williams, S. C., Koss, R. G., Tschurtz, B. A., Schmaltz, S. P., & Loeb, J. M. (2011). The perceived impact of public reporting hospital performance data: interviews with hospital staff. *International Journal for Quality in Health Care, 23*(6), 697–704.
- Han, A., & Lee, K.-H. (2021). *The impact of public reporting schemes and market competition on hospital efficiency*. Paper presented at the Healthcare.
- Han, A., & Park, J. (2021). Disparate impacts of two public reporting initiatives on clinical and perceived quality in healthcare. *Risk Management and Healthcare Policy, 5015–5025*.
- Hannan, E. L., Kumar, D., Racz, M., Siu, A. L., & Chassin, M. R. (1994). New York State’s cardiac surgery reporting system: four years later. *The Annals of thoracic surgery, 58*(6), 1852–1857.

- Hebden, J. N. (2012). Rationale for accuracy and consistency in applying standardized definitions for surveillance of health care - associated infections. *American Journal of Infection Control*, 40(5), S29-S31.
- Hibbard, J. H., Stockard, J., & Tusler, M. (2005). Hospital performance reports: impact on quality, market share, and reputation. *Health Affairs*, 24(4), 1150-1160.
- Hsieh, H.-F., & Shannon, S. E. (2005). Three approaches to qualitative content analysis. *Qualitative health research*, 15(9), 1277-1288.
- Hussey, P. S., Luft, H. S., & McNamara, P. (2014). Public reporting of provider performance at a crossroads in the United States: summary of current barriers and recommendations on how to move forward. *Medical Care Research and Review*, 71(5_suppl), 5S-16S.
- International Society for Pharmacoeconomics and Outcomes Research. (2016). ISPOR Strategic Plan. Retrieved from https://www.ispor.org/docs/default-source/about-ispor/ispor-strategic-plan-2016-2020-2018-06-05.pdf?sfvrsn=b2970b67_0
- Johnson, J. G., & Talbot, T. R. (2011). New approaches for influenza vaccination of healthcare workers. *Current opinion in infectious diseases*, 24(4), 363-369.
- Jung, J. K., Wu, B., Kim, H., & Polsky, D. (2016). The effect of publicized quality information on home health agency choice. *Medical Care Research and Review*, 73(6), 703-723.
- Jung, K., Shea, D., & Warner, C. (2010). Agency characteristics and changes in home health quality after Home Health Compare. *Journal of Aging and Health*, 22(4), 454-476.
- Konetzka, R. T., Brauner, D. J., Coca Perrailon, M., & Werner, R. M. (2015). The role of severe dementia in nursing home report cards. *Medical Care Research and Review*, 72(5), 562-579.
- Kumpunen, S., Trigg, L. J., & Rodrigues, R. (2014). *Public reporting in health and long-term care to facilitate provider choice*: Citeseer.
- Kurtzman, E. T., & Greene, J. (2016). Effective presentation of health care performance information for consumer decision making: a systematic review. *Patient Education and Counseling*, 99(1), 36-43.
- Lagu, T., & Greaves, F. (2015). From public to social reporting of hospital quality. *Journal of general internal medicine*, 30, 1397-1399.

- Lagu, T., Hannon, N. S., Rothberg, M. B., & Lindenauer, P. K. (2010). Patients' evaluations of health care providers in the era of social networking: an analysis of physician-rating websites. *J Gen Intern Med*, 25(9), 942-946. doi:10.1007/s11606-010-1383-0
- Lagu, T., Haskell, J., Cooper, E., Harris, D. A., Murray, A., & Gardner, R. L. (2019). Physician beliefs about online reporting of quality and experience data. *Journal of general internal medicine*, 34, 2542-2548.
- Lee, H. Y., Jin, S. W., Henning-Smith, C., Lee, J., & Lee, J. (2021). Role of health literacy in health-related information-seeking behavior online: cross-sectional study. *Journal of medical Internet research*, 23(1), e14088.
- Lihn, S. L., Kugler, J. D., Peterson, L. E., Lannon, C. M., Pickles, D., & Beekman III, R. H. (2015). Transparency in a pediatric quality improvement collaborative: a passionate journey by NPC - QIC clinicians and parents. *Congenital heart disease*, 10(6), 572-580.
- Lloyd, S., Cliff, C., FitzGerald, G., & Collie, J. (2021). Can publicly reported data be used to understand performance in an Australian rural hospital? *Health Information Management Journal*, 50(1-2), 35-46.
- Marcotte, L. M., Issaka, R. B., & Agrawal, N. (2021). Considerations of bias and reliability in publicly reported physician ratings. *Journal of general internal medicine*, 36(12), 3857-3858.
- Martin Emmert, & Wiener, M. (2017). What factors determine the intention to use hospital report cards? The perspectives of users and non-users. *Patient Education and Counseling*, 100(7), 1394-1401. doi:10.1016/j.pec.2017.01.021
- Martin, M., Zingg, W., Hansen, S., Gastmeier, P., Wu, A., Pittet, D., . . . group, P. s. (2013). Public reporting of healthcare-associated infection data in Europe. What are the views of infection prevention opinion leaders? *Journal of Hospital Infection*, 83(2), 94-98.
- Mazor, K. M., Dodd, K. S., & Kunches, L. (2009). Communicating hospital infection data to the public: a study of consumer responses and preferences. *American Journal of Medical Quality*, 24(2), 108-115.
- Mishra, N. K., Duke, J., Lenert, L., & Karki, S. (2020). Public health reporting and outbreak response: synergies with evolving clinical standards for interoperability. *Journal of the American Medical Informatics Association*, 27(7), 1136-1138.

- Natale, J., Joseph, J. G., Honomichl, R. D., Bazanni, L. G., Kagawa, K. J., & Marcin, J. P. (2011). Benchmarking, public reporting, and pay-for-performance: a mixed-methods survey of California pediatric intensive care unit medical directors. *Pediatr Crit Care Med*, 12(6), e225-e232.
- Negele, D., Lauerer, M., Nagel, E., & Ulrich, V. (2023). How to further develop quality competition in the German healthcare system? Results of a Delphi expert study. *Health Policy*, 138, 104937.
- Okoli, C. (2015). A guide to conducting a standalone systematic literature review. *Communications of the Association for Information Systems*, 37.
- Paddock, S. M. (2014). Statistical benchmarks for health care provider performance assessment: a comparison of standard approaches to a hierarchical Bayesian histogram - based method. *Health services research*, 49(3), 1056-1073.
- Paris, B., Arahood, T., Asche, C., & Amundson, G. (2013). Voluntary reporting of health care personnel seasonal influenza vaccination rates and the impact of universal policies in Illinois hospitals. *Vaccine*, 31(3), 514-517.
- Pimentel, C. B., Clark, V., Baughman, A. W., Berlowitz, D. R., Davila, H., Mills, W. L., . . . Hartmann, C. W. (2021). Health care providers and the public reporting of nursing home quality in the united states department of veterans affairs: protocol for a mixed methods pilot study. *JMIR research protocols*, 10(7), e23516.
- Poldrugovac, M., Amuah, J. E., Wei-Randall, H., Sidhom, P., Morris, K., Allin, S., . . . Kringos, D. (2022). Public Reporting of performance indicators in long-term care in Canada: does it make a difference? *Canadian Journal on Aging/La Revue canadienne du vieillissement*, 41(4), 565-576.
- Prang, K.-H., Canaway, R., Bismark, M., Dunt, D., Miller, J. A., & Kelaher, M. (2021). The impact of public performance reporting on cancer elective surgery waiting times: a data linkage study. *BMC Health Services Research*, 21, 1-10.
- Rambur, B., Vallett, C., Cohen, J. A., & Tarule, J. M. (2013). Metric-driven harm: an exploration of unintended consequences of performance measurement. *Applied Nursing Research*, 26(4), 269-272.
- Rhoads, K. F., Konety, B. M., & Dudley, R. A. (2009). Performance measurement, public reporting, and pay-for-performance. *Urologic Clinics of North America*, 36(1), 37-48.

- Saghafian, S., & Hopp, W. J. (2020). Can Public Reporting Cure Healthcare? The Role of Quality Transparency in Improving Patient - Provider Alignment. *Operations Research*, 68(1), 71-92.
doi:10.1287/opre.2019.1868
- Sander, U., Emmert, M., Dickel, J., Meszmer, N., & Kolb, B. (2015). Information presentation features and comprehensibility of hospital report cards: design analysis and online survey among users. *Journal of medical Internet research*, 17(3), e68.
- Schlesinger, M., Grob, R., & Shaller, D. (2015). Using patient - reported information to improve clinical practice. *Health services research*, 50, 2116-2154.
- Shaller, D., Kanouse, D. E., & Schlesinger, M. (2014). Context-based strategies for engaging consumers with public reports about health care providers. *Medical Care Research and Review*, 71(5_suppl), 17S-37S.
- Shintani Smith, S., Cheng, B. T., Kern, R. C., Cameron, K. A., & Micco, A. G. (2021). Publicly reported patient satisfaction scores in academic otolaryngology departments. *The Laryngoscope*, 131(10), 2204-2210.
- Slevin, P., Kessie, T., Cullen, J., Butler, M., Donnelly, S., & Caulfield, B. (2020). Exploring the barriers and facilitators for the use of digital health technologies for the management of COPD: a qualitative study of clinician perceptions. *QJM: An International Journal of Medicine*, 113(3), 163-172.
- Smith, M. A., Gigot, M., Harburn, A., Bednarz, L., Curtis, K., Mathew, J., & Farrar-Edwards, D. (2023). Insights into measuring health disparities using electronic health records from a statewide network of health systems: A case study. *Journal of clinical and translational science*, 7(1), e54.
- Snowden, A. M., Kunerth, V., Carlson, A. M., McRae, J. A., & Vetta, E. (2012). Addressing health care disparities using public reporting. *American Journal of Medical Quality*, 27(4), 275-281.
- Statista. (2022). Nominal GDP Driven by Digitally Transformed and Other Enterprises Worldwide 2018 - 2023. In.
- Tang, Y., Liu, C., & Zhang, X. (2017). Performance associated effect variations of public reporting in promoting antibiotic prescribing practice: a cluster randomized-controlled trial in primary healthcare settings. *Primary Health Care Research & Development*, 18(5), 482-491.

- Timofeyev, Y., Dremova, O., & Jakovljevic, M. (2023). The impact of transparency constraints on the efficiency of the Russian healthcare system: systematic literature review. *Journal of Medical Economics*, 26(1), 95–109.
- Totten, A. M., Wagner, J., Tiwari, A., O’Haire, C., Griffin, J., & Walker, M. (2012). Closing the quality gap: revisiting the state of the science (vol. 5: public reporting as a quality improvement strategy). *Evidence report/technology assessment*(208.5), 1–645.
- Toussaint, J., Shortell, S., & Mannon, M. (2014). *Improving the value of healthcare delivery using publicly available performance data in Wisconsin and California*. Paper presented at the Healthcare.
- Van Wilder, A., Brouwers, J., Cox, B., Bruyneel, L., De Ridder, D., Claessens, F., . . . Vanhaecht, K. (2021). A decade of commitment to hospital quality of care: overview of and perceptions on multicomponent quality improvement policies involving accreditation, public reporting, inspection and pay-for-performance. *BMC Health Services Research*, 21, 1–11.
- Vaughn, S. X., Maxey, H. L., Keen, A., Thoele, K., & Newhouse, R. (2020). Assessing public behavioral health services data: a mixed method analysis. *Substance Abuse Treatment, Prevention, and Policy*, 15, 1–10.
- Vom Brocke, J., Simons, A., Riemer, K., Niehaves, B., Plattfaut, R., & Cleven, A. (2015). Standing on the shoulders of giants: Challenges and recommendations of literature search in information systems research. *Communications of the Association for Information Systems*, 37(1), 9.
- Wadhera, R. K., O’ Brien, C. W., Joynt Maddox, K. E., Ho, K. K., Pinto, D. S., Resnic, F. S., . . . Yeh, R. W. (2019). Public reporting of percutaneous coronary intervention outcomes: institutional costs and physician burden. *Journal of the American College of Cardiology*, 73(20), 2604–2608.
- Walters, R. S., Albright, H. W., Weber, R. S., Feeley, T. W., Hanna, E. Y., Cantor, S. B., . . . Burke, T. W. (2014). Developing a system to track meaningful outcome measures in head and neck cancer treatment. *Head & Neck*, 36(2), 226–230.
- Wang, Y., Kung, L., & Byrd, T. A. (2018). Big data analytics: Understanding its capabilities and potential benefits for healthcare organizations. *Technological forecasting and social change*, 126, 3–13.
- Watson, J. W. R. T. (2002). Analyzing the Past to Prepare for the Future: Writing a Literature Review. *MIS Quarterly*, 26(2).

- Werner, R. M., & Asch, D. A. (2005). The unintended consequences of publicly reporting quality information. *Jama*, 293(10), 1239-1244.
- Werner, R. M., Konetzka, R. T., & Polsky, D. (2016). Changes in consumer demand following public reporting of summary quality ratings: an evaluation in nursing homes. *Health services research*, 51, 1291-1309.
- Werner, R. M., Konetzka, R. T., Stuart, E. A., Norton, E. C., Polsky, D., & Park, J. (2009). Impact of public reporting on quality of postacute care. *Health services research*, 44(4), 1169-1187.
- Yang, L., Liu, C., Wang, L., Yin, X., & Zhang, X. (2014). Public reporting improves antibiotic prescribing for upper respiratory tract infections in primary care: a matched-pair cluster-randomized trial in China. *Health research policy and systems*, 12, 1-9.
- Yonek, J. C., Mittler, J. N., & Hasnain-Wynia, R. (2014). Why and how six aligning forces for quality communities have focused on reducing disparities. *Medical Care Research and Review*, 71(5), 435-449.

Appendix A. Summary of the reviewed articles

No	Article	Country	Public reporting type	Data components	Research perspectives	Main finding
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1	Duncanson et al.(2024)	Australia	Performance Measurement/Registries	infection prevention scores/kidney disease quality data	data and quality metrics/presentation and information design	We suggest that additional data about centers and their patient cohorts be clearly explained alongside results to inform interpretations and judgements, such as centers' size, patient comorbidities, and socioeconomic status of the area. Adjusted outcomes with explanation of analysis may address this. Further research is needed to formally evaluate preferences for and comprehension of display options of comparative performance data among patients and how these may influence decisions about healthcare.
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2	Davila et al. (2024)	United States	Rating Systems	Medicare Nursing Home Compare	data and quality metrics	For the Inner Setting domain, respondents described organizational factors that both facilitated (staff motivation, teamwork, staff engagement) and impeded (staff turnover, perception that some quality measures are difficult to improve) their quality improvement efforts.
3	Smith et al.(2023)	United States	-	Electronic Health Records	data and quality metrics	Building on a statewide, centralized data repository of validated quality measures, we describe a process to select, assess, harmonize, standardize, and implement health disparities indicators to stratify and publicly report the quality measures. Building on a statewide, centralized data repository of validated quality measures, we describe a process to select,

						assess, harmonize, standardize, and implement health disparities indicators to stratify and publicly report the quality measures.
4	Negele et al. (2023)	Germany	-	nursing home report cards	Policy intervention	a step-by-step further development of the quality competitive instruments was favored. In this context, it was also important for the experts to have systematic and close-meshed scientific monitoring of the measures.
5	Bayindir and Schreyögg (2023)	Germany	-	Risk-Adjusted Outcome Indicators/quality indicators	Policy intervention	This supports the hypothesis that low-quality hospitals are more likely to engage in quality benchmarking, whereas for high-quality hospitals, preserving

						reputation and altruistic motives are also important.
6	Poldrugov ac et al. (2022)	Canada	Performance Measurement/Registr ies	CIHI	Policy intervention	Because of a “ceiling effect”, publication of performance data may have little effect on facilities that had already reached their peak performance and on indicators for which an improvement trend was present prior to publication.
7	Chae et al. (2022)	Korea	Monitoring and prevention Systems	Acute myocardial infarction	Policy intervention	Disparity exists in quality of care across hospitals, with tertiary care centers equipped with abundant resources offering better quality than general care centers. The variation in public reporting across individual hospitals underscores the need to investigate their

						management features and practice norms.
8	Van Wilder et al. (2021)	Belgium	-	-	Policy intervention	To assure quality of care remains a top-priority for acute-care hospitals, we recommend a revision of the current multicomponent quality policy where the adoption of all initiatives is streamlined and co-created bottom-up.

9	Prang et al. (2021)	Australia	Patient-Centered Systems	Elective surgery wait times	Policy intervention	Across all cancer types, urgent patients were all treated within 30 days before and after PPR. Following PPR, there was a slight increase in the mean waiting times across all cancer types and urgency categories. Patients with lung cancer waited on average two and half days longer for treatment and patients with breast cancer waited on average half-a-day less. There was no effect of PPR on waiting times for patients with bowel cancer across urgency categories.
10	Pimentel et al. (2021)	United States	-	Community Living Centers) care quality data	data and quality metrics	The knowledge gained from this pilot study and from future work will help V A refine how CLC Compare is used, ensure that CLC staff understand and are motivated to use its quality data, and implement

						concrete actions to improve clinical quality.
11	Lloyd et al. (2021)	Australia	-	-	data and quality metrics/presentation and information design	Visualization of multiple measures of performance over time was achievable but required a high degree of health information management skills. These are the timeliness of data, how peer groups are reported at different levels of granularity depending on the item reported, and the level of health information management skills required to consolidate and visualize data need to be sourced, combined and presented using visualization tools so that they can be interpreted

						and used by health service managers and clinicians.
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12	Han and Park (2021)	United States	-	-	Policy intervention	Despite the theoretically acknowledged merits of transparency, the present study noted that its ability to enhance quality of care remains inconclusive, and the pursuit of transparency may even inadvertently harm quality of care. ⁴⁰ The disclosure of hospital performance information was found to have the power to differentiate hospitals' clinical and perceived quality, but it failed to reach the desired outcomes without market pressure.
13	Han and Lee (2021)	United States	Claims-Based Systems	full-time equivalent staff/operating expenses	Policy intervention	while hospitals facing less competition responded significantly to increasingly transparent information by enhancing their efficiency relative to hospitals in more competitive markets.

14	Vaughn et al. (2020)	United States	Hospital and institutional systems	public database systems (e.g., databases maintained by SAMHSA)	data and quality metrics	The results of this study highlight the need for comprehensive and timely information about treatment facilities which can be used for clinical decision-making, facilitating care coordination and assessing organizational effectiveness. Data not typically captured by public sources, such as wait times, intake procedures and current capacity, were helpful for identifying factors which can inform referrals to behavioral health treatment.
15	Saghafian and Hopp (2020)	United States	Hospital and institutional systems	Data from Centers for Medicare and Medicaid Services	Policy intervention	Hence, a policymaker concerned with societal outcomes should accompany increasing quality transparency with other policies that correct the allocation of patients to hospitals.

16	Mishra et al. (2020)	United States	Monitoring and prevention Systems	-	data and quality metrics	Without semantic interoperability, data collected from various sources is not easily normalized for population-level surveillance and response.
17	Ellis et al. (2020)	United States	-	patient quality indicators	presentation and information design	Patients valued hospital reputation, physician experience, and primary care recommendations while publicly reported metrics like patient outcomes were less important. Public quality reports contain information that patients perceive to be of relatively low value, which may contribute to low utilization.

18	Wadhera et al. (2019)	United States	-	percutaneous Coronary Intervention	data and quality metrics	Given that reporting has not been shown to improve care or outcomes, and may instead incentivize physician risk aversion, these data raise concern about the value of current efforts to publicly report PCI outcomes. As public reporting continues to expand nationally under the ACC, it will be important to ensure these efforts are efficient and do not result in excessive cost and administrative burden to physicians and institutions.
19	Lagu et al. (2019)	United States	Comparative Reporting Systems	patient experience data	data and quality metrics	physicians are unaware of mechanisms for publicly reporting quality data and doubt the accuracy of information about physicians that is present online.

20	Geraedts et al. (2018)	Germany	Public reporting systems	physician information	presentation and information design	Because patients' and physicians' importance rankings of different criteria do not match, report cards should consider these preferences in order to increase attractiveness of the report cards for different audiences. Our results indicate that hospital report cards for referring physicians counselling patients on their choice of hospital should highlight a feature that today's reports miss: information on physicians' experiences with hospitals.
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21	Tang et al. (2017)	China	Patient-Centered Systems	patient demographics/30-day readmission rates	Policy intervention	In a qualitative study, the investigated primary practitioners expressed their concerns that public reporting, which encourages the ‘name and shame’ culture, would exert stress on them (Marshall et al., 2002). This pressure, both from the political concerns and reputation, triggers change in behaviour and prescription patterns in this study, especially in low performers
22	Emmert and Wiener (2017)	Germany	Public reporting systems	Hospital Report Cards	presentation and information design	An alternative and partly related explanation may be that HRC present hospital-quality information in a way that is too complex for ‘ordinary’ users. Especially patients who suffer from severe health problems can clearly benefit from the increased transparency provided by HRC.

23	Emmert et al. (2017)	United States/Germany	Public reporting systems	patient satisfaction scores	presentation and information design	As shown in the US, Consumer Reports and Health-Grades handle missing data by establishing minimum data thresholds for a hospital to receive a rating. If the data threshold is not met, a rating is not calculated. This mechanism is also applied on the German report card AOK-Krankenhausnavigator.
24	Werner et al. (2016)	United States	Rating Systems	five-star rating data	presentation and information design	These results support the use of summary measures in report cards.
25	Jung et al. (2016)	United States	-	-	Policy intervention	Identifying and releasing quality information that is meaningful to consumers may help increase consumers' use of public reports.
26	Fakih et al. (2016)	United States	Public reporting systems	Healthcare-associated infections (HAIs) data	Policy intervention	The ICSib provides information on how a facility fares compared with others during the same period of time. This helps health systems and quality collaboratives to identify

						those facilities that are performing at a less than optimal level compared with others.
27	Crews et al. (2016)	United States	Claims-Based Systems	mortality rates	data and quality metrics	process for claims-based measures like PSIs raise concerns about the use of these data to make meaningful judgments about quality, safety, and value. The authors recommend a standardized approach to PSI data reporting and validation be designed and implemented in the near future.
28	Zhang et al. (2015)	China	-	prescription quality indicators	presentation and information design	increasing the accessibility of prescription quality indicators can create a positive attitude on transparency regulation, which would increase the intention of GPs to improve prescribing practices.

29	Schlesinger et al. (2015)	United States/Europe	-	Patient-Reported information	presentation and information design	First, we identify four forms of “patient-reported information” (PRI), each with distinctive roles shaping clinical practice: (1) patient-reported outcomes measuring self-assessed physical and mental well-being, (2) surveys of patient experience with clinicians and staff, (3) narrative accounts describing encounters with clinicians in patients’ own words, and (4) complaints/grievances signaling patients’ distress when treatment or outcomes fall short of expectations.
30	Sander et al. (2015)	Germany	Public reporting systems	-	presentation and information design	Best-practice evidence identified in international literature was in agreement with 5 findings about German report card designs: (1) avoid tables without symbols, (2) include bar charts with symbols, (3) state explicitly whether high or low values indicate good

						performance or provide a “good quality” range, (4) avoid incomplete data (N/A given as a value), and (5) rank hospitals by performance.
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31	Lihn et al. (2015)	United States	-	-	data and quality metrics	Activities are organized around four interrelated levels of transparency (individual, organizational, collaborative, and system), each with a specified purpose and aim. A number of Transparency Work Group recommendations have been operationalized. Aggregate collaborative performance is now reported on the public-facing web site. Specific information that the Transparency Work Group recommends centers provide to parents has been developed and published. Obtain Transparency Report Card data on a semiannual basis to monitor each center's progress. Share these data with all NPC-QIC centers transparently. Revise the collaborative's Data Use Agreement to allow for transparent
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						sharing of data between participating care centers.
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32	Lagu and Greaves (2015)	United States	Public reporting systems	-	presentation and information design	Anecdotal evidence from other social media websites such as Twitter have shown that the hospitals most adept at engaging patients on social media are hospitals that are well known for providing high-quality care (e.g., the Cleveland Clinic has more than a third of a million followers; the Mayo Clinic has more than a million).
33	Konetzka et al. (2015)	United States	Public reporting systems	nursing home report cards	Policy intervention	We see no evidence that individuals with severe dementia are being avoided by nursing homes, despite their potential negative impact on quality scores, but we do find an increase in coding of end-stage disease.
34	Du et al. (2015)	China	Hospital and institutional systems	prescription quality indicators/Antibiotic prescription percentage ranking/prescription behavior	Policy intervention	This study protocol outlines a design that aims to examine the transparency mechanism and to evaluate the effect of public reporting on prescription.

				data/prescription quality indicator data		
35	Yonek et al. (2014)	United States	-	chronic condition management	data and quality metrics	We conclude that community partnerships are a promising mechanism for targeting efforts to increase awareness of public reports in minority communities.
36	Yang et al. (2014)	China	Hospital and institutional systems	-	data and quality metrics	A high quality information system is also critical for easy access and quality assurance of data.

37	Walters et al. (2014)	United States	Patient-Centered Systems	-	data and quality metrics	It clearly defines the areas that need to be improved upon if we are to use this data to improve care and demonstrate value on an ongoing basis. If we develop quality systems that utilize patient-centered outcome metrics that are documented clearly in the course of treatment and can be periodically reported electronically, we will have made a great advance in the development of a true value-based cancer care delivery system. This study also serves to define elements for the design of information systems so relevant data can be captured prospectively using structured clinical documentation.
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38	Toussaint et al. (2014)	United States	Claims-Based Systems	publicly available performance data/Publicly available claims data	data and quality metrics	The Wisconsin and California experiences suggest that the cost and quality of healthcare can be improved dramatically in some cases and incrementally but steadily in other cases by making available accurate, actionable data.
39	Shaller et al. (2014)	United States	-	chronic condition management	presentation and information design	Cross-cutting lessons for report sponsors and policy makers include new media strategies such as a commitment to adaptive web-based reporting, new metrics with richer emotional content, and the use of navigators or advocates to assist consumers with interpreting reports. Our central contention in this article, however, is that the most fundamental impediments involve the challenges individual consumers face as they try to interpret this

						information and extract its relevance for their personal circumstances.
40	Paddock (2014)	United States	-	-	Policy intervention	Given the rewards and penalties associated with characterizing top performance, the ability of statistical benchmarks to summarize key features of the provider performance distribution should be examined.

41	Dehmer et al. (2014)	United States	-	-	presentation and information design	cardiovascular clinicians and professional societies should take a leadership role to: 1) continue the development of meaningful performance measures;
42	Hussey et al. (2014)	United States	-	-	presentation and information design/data and quality metrics	The recommendations include increased consumer engagement in report development, standardization of performance measures, and exploration of alternative technologies to target and deliver reports to consumers.
43	Christianson et al. (2014)	United States	Performance Measurement/Registries	-	presentation and information design	In our study, respondents who were highly educated and in poorer health were more likely to be aware of PQL.
44	Rambur et al. (2013)	United States	Performance Measurement/Registries	quality indicators	data and quality metrics	to realize a vision of patient-centered, context sensitive care, nurses must truly be equal partners in clinical decision making.

45	Paris et al. (2013)	Europe	Monitoring and prevention Systems	flu vaccination rates (staff behavior data)	Policy intervention	There is growing evidence that implementation of a universal policy contributes to a dramatic increase in performance.
46	Martin et al. (2013)	Europe	Comparative Reporting Systems	prescriber details (name, work unit, time of prescriptions)	Policy intervention	Finally, the majority of the experts acknowledged the positive influence on hospitals by increasing competition on the basis of quality, but they are hesitant about publishing infection rates as these can be misinterpreted by patients and need standardization and validation.
47	Emmert et al. (2013)	Germany	Rating Systems	User-generated data	presentation and information design	Neither health policy makers nor physicians should underestimate the influence of physician-rating websites. They already play an important role in providing information to help patients decide on an appropriate physician. This is especially true of females with a higher education status,

						who are insured by SHI, and who utilize the health care sector at a higher rate.
48	Emmert and Meier (2013)	Germany	Rating Systems	-	presentation and information design	An increase in the usage of PRWs might contribute to reducing the lack of publicly available information on physician quality.

49	Snowden et al. (2012)	United States	Monitoring and prevention Systems	chronic condition management	data and quality metrics	Improving performance rates for all patients requires a thorough examination of performance data at a medical group or clinic level in addition to data at national, state-wide, and health plan levels. The public reporting of healthcare performance rates can serve as an impetus to quality improvement, particularly when rates are reported at a medical group level or eventually at the clinic level where more accountability lies and actions to address disparities are more likely.
50	Hebden (2012)	United States	-	Healthcare-associated infections (HAIs) data	data and quality metrics	Efforts to improve data reliability include CDC funding support of automated infection detection processes and state validation efforts, continual improvements in the NHSN application, and intensive educational

						offerings for IPs performing surveillance.
51	Natale et al. (2011)	United States	-	Pediatric Intensive Care Unit performance data	data and quality metrics	Our data also highlight the importance of including PICU leadership in designing data collection and reporting systems for the purpose of quality improvement.
52	Johnson and Talbot (2011)	United States	-	Healthcare worker (HCW) flu vaccination coverage/prescriber details (name, work unit, time of prescriptions)	Policy intervention	Public reporting of institutional HCW influenza vaccination rates is another strategy to achieve high vaccination rates, as HCW influenza vaccination may be used in the future as a quality and safety metric.

53	Hafner et al. (2011)	United States	-	-	Policy intervention	Interviewees from hospitals with lower measure rates were more concerned about the timeliness of the data, and they expressed concerns regarding the methodological accuracy of data collection, whereas interviewees from higher performing hospitals tended to view data quality challenges as learning opportunities.
54	Jung et al. (2010)	United States	-	HHC (Home Health Compare) public reporting data	Policy intervention	There was large variation in the degree of quality improvement after HHC by quality indicators and by agency characteristics.
55	Christianson et al. (2010)	United States	Comparative Reporting Systems	-	presentation and information design	Purchasers and consumers value data pertaining to individual physicians, but many report sponsors may not possess sufficient data to construct reliable, representative measures (especially relating to treatment of specific diseases) at this

						level.25 Current approaches to addressing this problem need further development.
56	Werner et al. (2009)	United States	-	Care website (accessstocare.va.gov)	Policy intervention	One possible way to do this is to combine public reporting with pay for performance.
57	Rhoads et al. (2009)	United States	Monitoring and prevention Systems	-	Policy intervention	Input from urologists in the development of PFP programs is crucial in determining the most effective ways to administer financial incentives.

58	Mazor et al. (2009)	United States	Monitoring and prevention Systems	Healthcare-associated infections (HAIs) data	presentation and information design	it does suggest that the general public might find materials such as these less understandable than the respondents in our sample found them. Clearly, the educational level of the intended audience should be considered when developing materials for consumers.
59	Edmond and Bearman (2007)	United States	-	Healthcare-associated infections (HAIs) data	data and quality metrics	To date, legislation has been devolved to individual states, resulting in multiple rules, standards, and methodology, which precludes the ability to compare hospitals nationally.
60	Werner et al. (2005)	United States	-	Hospital Report Cards	Policy intervention	Finally, if public report cards are to improve the quality of care, participation must be mandatory and quality measurement and reporting must be universally adopted. Otherwise, providers who receive low-

						quality scores face incentives to avoid reporting and the sickest patients will be shifted from rated to unrated providers.
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